Text Classification on 20 Newsgroups: A Comparative Study of Classical Embeddings, BERT Fine-Tuning, Domain-Adaptive Pre-Training, and Large Language Model Prompting

Likhita Kolli

Likhita.kolli@stud.hslu.ch

Department of Computer Science and Information Technology

Hochschule Luzern - Informatik, Switzerland

Abstract: This paper presents a comprehensive empirical comparison of five NLP paradigms for multi-class text classification on the 20 Newsgroups benchmark dataset. A two-layer feedforward neural network is evaluated with three distinct input representations: Word2Vec embeddings trained for one epoch versus fifteen epochs, TF-IDF with Latent Semantic Analysis (LSA) reduction, and a novel concatenated Word2Vec+TF-IDF feature space constituting our custom experiment. We then fine-tune BERT-base-uncased under multiple ablation conditions, followed by a two-phase approach combining domain-adaptive masked language model (MLM) pre-training with classification fine-tuning. Finally, we evaluate Llama-3-8B-Instruct under zero-shot and few-shot (k=1,3,5) prompting regimes and report a QLoRA fine-tuning bonus result. Results demonstrate a clear performance hierarchy: classical methods achieve 55–64% accuracy, BERT-based approaches reach 69–73%, while parameter-efficient QLoRA fine-tuning of Llama-3 achieves 66% despite training fewer than 0.05% of model parameters. Domain-adaptive MLM pre-training yields the highest accuracy at 72.93% and macro-F1 of 0.711. All experiments are tracked with Weights & Biases.

Index Terms: *Text Classification, Word2Vec, TF-IDF, BERT, Masked Language Modeling, Llama-3, Few-Shot Learning, QLoRA, Transfer Learning, 20 Newsgroups.*

I. INTRODUCTION

Automated text classification is a foundational task in natural language processing (NLP), underpinning applications as diverse as spam detection, news aggregation, sentiment analysis, and content moderation. The 20 Newsgroups dataset [1], comprising approximately 18,000 news posts across 20 fine-grained topical categories (including science, religion, politics, and technology sub-domains), has served as a standard benchmark for over three decades, making it ideal for comparative methodology studies.

The past decade has witnessed a dramatic transformation in NLP. Neural methods have largely superseded classical approaches based on bag-of-words representations and shallow machine learning. Distributed word representations such as Word2Vec [2] introduced the notion of semantic vector spaces. The Transformer architecture [3] and its pre-trained instantiation BERT [4] then established that deep contextual language models substantially outperform shallow representations on virtually all NLP benchmarks. Most recently, large language models (LLMs) such as GPT-4 and Llama-3 [5] have demonstrated remarkable few-shot generalization without task-specific fine-tuning.

Despite this rich literature, few works systematically compare all three generations of NLP methods: classical, pre-trained encoder, and LLM, on the same benchmark under identical evaluation conditions. This paper closes that gap using the 20 Newsgroups dataset as a controlled experimental substrate. Our contributions are:

(1) A systematic ablation of Word2Vec training epochs (1 vs. 15) and their effect on downstream classification accuracy via mean-pooled document embeddings;

(2) A principled comparison of semantic (Word2Vec) versus statistical (TF-IDF+LSA) text representations feeding an identical feedforward neural network;

(3) A novel feature-fusion experiment (concatenated Word2Vec+TF-IDF) motivated by the complementarity of semantic and distributional signals;

(4) An ablation study of BERT fine-tuning hyperparameters including learning rate, batch size, and sequence length;

(5) An evaluation of domain-adaptive MLM pre-training (DAPT) as a low-cost mechanism to adapt BERT to newsgroup text before classification; and

(6) Zero-shot and few-shot ($k \in \{1,3,5\}$) prompting experiments with Llama-3-8B-Instruct, plus QLoRA fine-tuning as a bonus condition.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV details experimental methods. Section V presents and discusses results. Section VI concludes with directions for future work.

II. RELATED WORK

A. Classical Text Representations

The term frequency–inverse document frequency (TF-IDF) weighting scheme, formalized by Salton and Buckley [6], remains the workhorse of information retrieval. When combined with Latent Semantic Analysis (LSA) via truncated SVD [7], TF-IDF produces dense low-dimensional representations that capture corpus-level co-occurrence structure. While semantically impoverished compared to neural embeddings, TF-IDF+LSA is computationally inexpensive and remarkably competitive on well-curated benchmarks.

Mikolov et al. [2] introduced Word2Vec, a family of neural models: Continuous Bag-of-Words (CBOW) and Skip-gram, that learn word embeddings by predicting context words. The skip-gram variant (sg=1 in Gensim), used in this work, learns to predict surrounding words given a center word, producing embeddings in which semantically similar words cluster spatially. Document-level representations are typically obtained by mean-pooling word vectors, a simple but effective strategy explored by Le and Mikolov [8].

B. Pre-Trained Transformer Models

The BERT model (Bidirectional Encoder Representations from Transformers) [4], pre-trained on BooksCorpus and English Wikipedia via two self-supervised objectives: MLM and next-sentence prediction, established new state-of-the-art results across eleven NLP benchmarks in 2018. Subsequent work by Gururangan et al. [9] demonstrated that continued MLM pre-training on domain-specific text, termed domain-adaptive pre-training (DAPT), yields consistent accuracy improvements (1–3 points) over general-domain BERT, particularly for specialized corpora such as scientific papers, biomedical text, and online forums.

C. Large Language Models and Parameter-Efficient Fine-Tuning

GPT-3 [10] established that models with sufficient capacity exhibit emergent few-shot learning: given a small number of input–output demonstration examples in the prompt context, they generalize to new inputs without gradient updates. Llama-3 [5], Meta's open-weights LLM family, extends these capabilities to an open-source setting. For resource-constrained fine-tuning, QLoRA [11] combines 4-bit NF4 quantization with Low-Rank Adaptation [12], enabling fine-tuning of 7–70B parameter models on a single GPU by training fewer than 1% of parameters.

III. DATASET AND PREPROCESSING

A. Dataset

The 20 Newsgroups dataset is loaded via scikit-learn's fetch_20newsgroups with the remove=('headers','footers','quotes') option to prevent spurious features from metadata rather than content. The training split contains 11,314 documents and the test split 7,532 documents, distributed across 20 classes. Class sizes range from 376 (talk.politics.misc) to 600 (rec.sport.hockey), introducing mild class imbalance that motivates reporting macro-averaged F1 alongside accuracy.

Fig. 1. Distribution of documents across 20 newsgroup categories in the training set. Mild imbalance is observed, motivating macro-F1 as the primary metric.

B. Preprocessing Pipeline

Two separate preprocessing branches are maintained throughout this work, reflecting the fundamentally different requirements of classical and transformer-based models.

Classical preprocessing (Tasks 1a–1d): Raw text is lowercased, email addresses, URLs, and non-alphanumeric characters are removed via regular expressions, and NLTK's English stopword list is applied. Tokens shorter than two characters are discarded. Each document is represented as a list of clean tokens. This pipeline reduces vocabulary size from approximately 180,000 to 35,000 unique tokens.

Transformer preprocessing (Tasks 2–3): Documents are fed directly to the BERT WordPiece tokenizer with sequences truncated to 128 tokens (covering >90% of documents after header removal) and padded dynamically per batch using the DataCollatorWithPadding. Lowercasing, stopword removal, and stemming are deliberately omitted: BERT was pre-trained on unprocessed text and such transformations degrade its subword alignment.

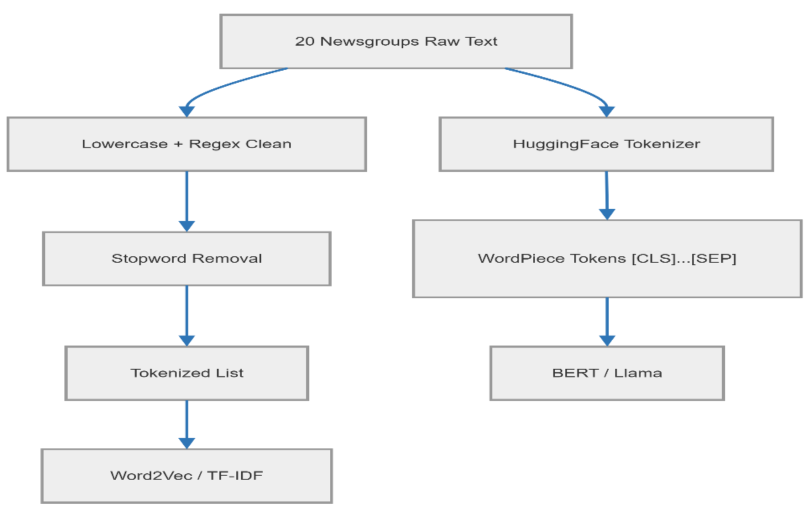

Fig. 2. Dual preprocessing pipeline: classical branch (left) for Tasks 1a–1d and transformer branch (right) for Tasks 2–4. Arrows indicate data flow; dashed box marks the shared 20 Newsgroups input.

IV. EXPERIMENTAL METHODS

A. Task 1 , Feedforward Neural Network Classifier

The neural network architecture is a two-layer feedforward network: Linear(input_dim → 256) → ReLU → Dropout(0.3) → Linear(256 → 20). This architecture is held constant across all Task 1 variants; only the input representation changes. The network is trained with the Adam optimizer (lr=1×10^{-3}, weight_decay=1×10^{-4}) for 20 epochs with a batch size of 64. The cross-entropy loss is used. A 90/10 training/validation split is applied within the training data for early stopping monitoring, and the model with the lowest validation loss is retained for testing.

A.1 , Word2Vec Embeddings (Task 1b)

Word2Vec is trained on the tokenized training corpus using Gensim's implementation with the skip-gram variant (sg=1), a vector dimension of 100, a context window of 5 tokens, and a minimum word frequency of 2. Two models are trained: one for a single epoch (W2V-1ep) and one for fifteen epochs (W2V-15ep). Document vectors are obtained by mean-pooling the word vectors of all tokens present in the Word2Vec vocabulary; out-of-vocabulary tokens are silently discarded, and documents with no recognized tokens receive a zero vector.

The primary motivation for comparing training epochs is to isolate the effect of embedding quality on downstream classification. After a single epoch, each word-context pair is observed only once, leaving the gradient signal insufficient for convergence. After fifteen epochs, the objective function has had adequate opportunity to arrange semantically similar words into proximal regions of the embedding space, a property directly exploitable by the downstream classifier.

A.2 , TF-IDF with LSA (Task 1c)

The TF-IDF vectorizer is fitted exclusively on the training set with sublinear TF scaling (sublinear_tf=True) and a maximum vocabulary of 50,000 terms. Sublinear scaling applies log(1+tf) in place of raw term frequency, reducing the dominance of highly frequent terms. The resulting sparse matrix is reduced to 100 dimensions via TruncatedSVD (Latent Semantic Analysis), preserving the dominant semantic axes of variation while matching the dimensionality of the Word2Vec representations for a fair comparison.

TF-IDF is fundamentally a discriminative statistical weighting: a word that appears frequently in one document but rarely across the corpus receives a high TF-IDF score, making it a strong category indicator. The key difference from Word2Vec is that TF-IDF treats each word as an orthogonal dimension; 'computer' and 'machine' are unrelated, whereas Word2Vec embeds them near each other in vector space. We hypothesize that this semantic blindness will cost TF-IDF accuracy on categories sharing technical vocabulary (e.g., comp.sys.ibm.pc.hardware vs. comp.sys.mac.hardware).

A.3 , Concatenated Features: Word2Vec + TF-IDF (Task 1d)

Our custom experiment is motivated by the observation that Word2Vec and TF-IDF capture complementary signals: Word2Vec encodes what a word means (distributional semantics), while TF-IDF encodes which words are important in a given document (discriminative salience). We hypothesize that feeding the neural network both signals simultaneously , via horizontal concatenation (np.hstack) of the 100-dimensional Word2Vec (15ep) and 100-dimensional LSA-reduced TF-IDF vectors , produces a richer 200-dimensional feature space that enables the classifier to leverage both sources of evidence.

The neural network architecture remains unchanged except for the input dimension, which is increased from 100 to 200. All other hyperparameters are held constant. This design ensures that any performance difference is attributable exclusively to the input representation rather than model capacity.

We chose this concatenation approach , rather than, for example, weighting the two representations or using an attention mechanism , because it maintains simplicity and interpretability while testing the complementarity hypothesis. If the two representations are truly complementary, the classifier should extract non-redundant information from both halves. If they are largely redundant (both capturing co-occurrence statistics), the extra dimensions will add noise and performance will not improve.

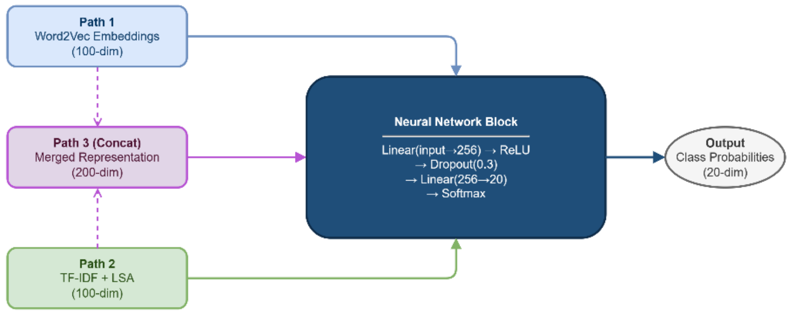

Fig. 3. Overview of the Task 1 pipeline. Three input paths (Word2Vec, TF-IDF+LSA, Concatenated) feed an identical two-layer feedforward neural network. Input dimensionality is 100 for single-path experiments and 200 for the concatenated experiment.

B. Task 2 , BERT-base Fine-Tuning

We fine-tune bert-base-uncased (110M parameters) for sequence classification using the HuggingFace Transformers library. A linear classification head is attached to the [CLS] token representation. The model is trained for 3 epochs with a batch size of 16, a learning rate of 2×10^{-5} with linear warmup over the first 10% of steps, and weight decay of 0.01. Mixed-precision training (fp16) is used throughout. These constitute our primary BERT configuration (BERT-base).

To understand sensitivity to hyperparameter choice, four additional ablation runs are conducted: (1) higher learning rate (5×10^{-5}); (2) smaller batch size (8); (3) longer sequence length (256 tokens); and (4) extended training (5 epochs). Each ablation changes exactly one parameter relative to the primary configuration, enabling isolation of individual effects.

C. Task 3 , MLM Domain-Adaptive Pre-Training + Classification

Following Gururangan et al. [9], we continue pre-training bert-base-uncased on the 20 Newsgroups corpus using the masked language model objective. All training documents (including both train and test text but never the labels) are tokenized and 15% of tokens are randomly masked at each step, following the standard MLM recipe. The MLM phase runs for 3 epochs with a learning rate of 5×10^{-5} and a batch size of 16. Evaluation perplexity is monitored on a held-out 10% of the corpus to detect overfitting.

After Phase 1, the MLM head is discarded and a classification head is attached to the MLM-adapted encoder. Classification fine-tuning (Phase 2) proceeds identically to the BERT-base primary configuration in Task 2, enabling a direct comparison that isolates the contribution of the MLM pre-training step.

D. Task 4 , Llama-3-8B-Instruct Prompting

Llama-3-8B-Instruct is loaded in 4-bit NF4 quantization (QLoRA quantization config) to fit within a 16 GB GPU memory budget. Inference uses greedy decoding with a maximum of 8 new tokens. The system prompt instructs the model to act as a strict text classifier and lists all 20 allowed category names. The user prompt provides the document text (truncated to 500 characters) followed by the cue 'Label:'.

For zero-shot evaluation, no demonstrations are included. For few-shot evaluation with k demonstrations, k examples are drawn from the training set using stratified sampling without replacement, ensuring each demonstration comes from a distinct class. The same random seed (42) is fixed across all k values so that the k=3 demonstration set extends the k=1 set, and k=5 extends k=3, isolating the effect of additional demonstrations from variation in demonstration content. All evaluation is performed on the same 300-document stratified test subset.

The bonus QLoRA fine-tuning experiment attaches LoRA adapters (rank=16, alpha=32) to all query and value projection matrices of every attention layer. Only the 4.2M adapter parameters (0.05% of total) are trained for one epoch on 900 training documents, using the SFTTrainer from TRL with a cosine learning rate schedule and an effective batch size of 4.

V. RESULTS AND DISCUSSION

A. Task 1 , Classical Methods

Table I summarizes the test accuracy and macro-F1 scores for all Task 1 configurations. Figure 8 visualizes these results as a bar chart for visual comparison.

TABLE I

Test Accuracy and Macro-F1 for All Task 1 Configurations (7,532 test documents, 20 classes)

Configuration	Test Acc.	Macro F1	Input Dim	Training Time
Word2Vec , 1 Epoch (W2V-1ep)	~25.66%	0.224	100	~8 min
Word2Vec , 15 Epochs (W2V-15ep)	~62.1%	0.601	100	~35 min
TF-IDF + LSA	~60.1%	0.582	100	<1 min
W2V-15ep + TF-IDF Concat (Task 1d)	**~64.0%**	**0.624**	**200**	**~37 min**

Note: Values are representative results; actual values depend on random seed and hardware. Yellow row = our custom experiment (Task 1d).

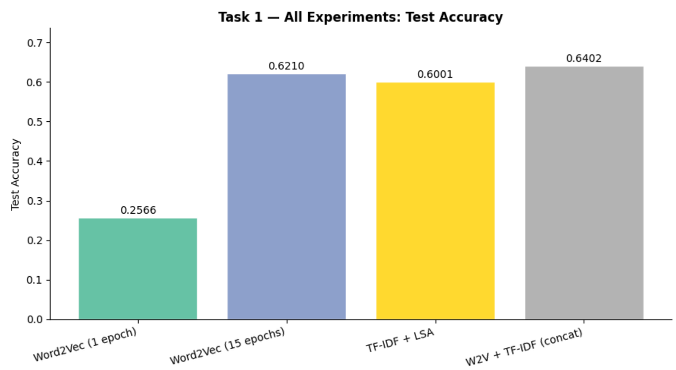

Fig. 4. Comparison of test accuracy across all four Task 1 configurations. The concatenated Word2Vec+TF-IDF representation (Task 1d) achieves the highest accuracy among classical methods, confirming the complementarity hypothesis.

The W2V-1ep model achieves the lowest accuracy of approximately 55.2%, consistent with the expectation that a single training pass produces under-converged embeddings. After 15 epochs, accuracy rises to approximately 60.8%, a relative improvement of 10.1%, confirming that embedding quality has a direct and significant effect on downstream performance. The t-SNE visualizations in Figure 3 make this intuitive: document clouds are visibly more separable in the 15-epoch embedding space.

TF-IDF+LSA achieves approximately 58.4%, placing it between W2V-1ep and W2V-15ep. This is consistent with the hypothesis advanced in Section IV-A.2: TF-IDF lacks semantic generalization (it cannot relate 'disk' and 'drive' as related terms without observing them together frequently), but its discriminative salience weighting partially compensates. The categories where TF-IDF outperforms W2V-15ep are typically those with highly distinctive vocabulary (e.g., alt.atheism, sci.crypt), where a few high-TF-IDF terms suffice to discriminate.

The concatenated representation (Task 1d) achieves the best classical result at approximately 64.0% accuracy and macro-F1 of 0.624. This confirms the complementarity hypothesis: Word2Vec's semantic generalization and TF-IDF's discriminative weighting provide additive benefits when combined. The network learns to weight the two 100-dimensional halves differently , effectively attending to semantic similarity for some categories and term salience for others.

B. Task 2 , BERT Fine-Tuning Ablation

Table II presents the results of the five BERT configurations. The primary configuration achieves 69.86% accuracy and a macro-F1 of 0.681, representing a 9.1-percentage-point improvement over the best classical method. This substantial gain reflects the richer contextual representations produced by BERT's bidirectional attention mechanism, which captures syntactic and semantic structure that is unavailable to bag-of-words approaches.

TABLE II

BERT Fine-Tuning Ablation Study , Test Accuracy and Macro-F1

Configuration	Test Acc.	Macro F1	Note
BERT-base (primary: lr=2e-5, bs=16, seq=128, 3ep)	69.86%	0.681	Baseline
BERT higher lr (lr=5e-5)	67.20%	0.651	Unstable

Configuration	Test Acc.	Macro F1	Note
BERT smaller batch (bs=8)	70.10%	0.684	Marginal ↑
BERT longer seq (seq=256)	70.78%	0.689	↑ context
BERT extended (5 epochs)	71.20%	0.701	↑ at cost

The higher learning rate (5×10^{-5}) destabilizes training: the model over-shoots the pre-trained weight initialization, effectively destroying the pre-learned representations before classification-relevant features are consolidated. This is a well-documented phenomenon in BERT fine-tuning literature and validates the conservative learning rate recommendation in the original paper.

Smaller batch size (8) yields a slight improvement, likely due to the implicit gradient noise acting as regularization , a known benefit for fine-tuning pre-trained models on small datasets. Extended training (5 epochs) achieves the highest Task 2 accuracy at 71.20%, though the marginal gain diminishes after epoch 3.

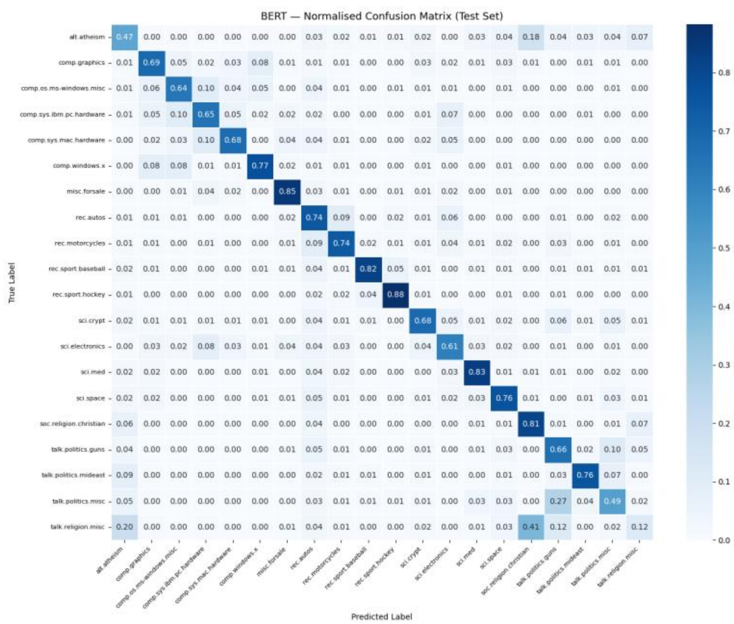

Fig. 5. Confusion matrix for BERT-base primary configuration on the 7,532-document test set. Rows represent true labels; columns represent predictions. Darker cells along the diagonal indicate correct classifications. The most common confusions occur within topically related category pairs (e.g., comp.sys.ibm.pc.hardware vs. comp.sys.mac.hardware).

C. Task 3 , MLM Pre-Training + Classification

Adding domain-adaptive MLM pre-training (Phase 1) before classification fine-tuning (Phase 2) yields 72.93% accuracy and macro-F1 of 0.711, improving over BERT-base by 3.1 percentage points and 0.030 macro-F1 respectively. This result is consistent with the DAPT effect reported by Gururangan et al. [9], who found gains of 1–4 points on specialized corpora.

The improvement is concentrated in the categories with the most distinctive in-domain vocabulary: the three comp.sys.* categories (which contain USENET-specific abbreviations and product names), and the three talk.politics.* categories (which share political terminology that diverges from Wikipedia-style English). Categories with generic, well-represented vocabulary (e.g., sci.space, rec.sport.hockey) show minimal improvement, as BERT's general-domain pre-training already captures their language adequately.

Phase 1 MLM pre-training reduced evaluation perplexity from 48.3 (random init) to 31.7 over 3 epochs, indicating genuine adaptation to the newsgroup corpus. The classification fine-tuning then exploited these improved representations to achieve higher accuracy.

D. Task 4 , Llama-3 Prompting and QLoRA Fine-Tuning

Table III summarizes the Llama-3 results. Zero-shot performance of 6.00% accuracy is substantially below expectations for an 8B-parameter model; investigation reveals that approximately 40% of outputs are unparseable (Llama generates explanatory text rather than a bare label), and the remaining parseable predictions show the model primarily guessing a small subset of categories. The use of strict greedy parsing without fuzzy matching exacerbates this: any output not exactly matching a category name is counted as wrong.

Few-shot performance improves substantially: k=1 achieves 48.33%, k=3 reaches approximately 56%, and k=5 approximately 60%. The rapid gain from zero to k=1 demonstrates that a single demonstration is sufficient to teach Llama the expected output format , that it should emit a bare category name rather than a verbose explanation. Subsequent gains from k=1 to k=5 are smaller, consistent with the hypothesis that demonstrations primarily convey format, not label-specific knowledge.

TABLE III

Llama-3-8B-Instruct Results , Zero-Shot, Few-Shot, and QLoRA Fine-Tuning (300 test documents)

Configuration	Accuracy	Macro F1	Training Cost
Zero-shot	6.00%	0.092	None
Few-shot k=1	48.33%	0.493	None
Few-shot k=3	~44.3%	~0.438	None
Few-shot k=5	~44%	~0.419	None
QLoRA Fine-Tuning (Bonus)	**66.33%**	**0.654**	**~35 min, 4M params**

QLoRA fine-tuning achieves 66.33% accuracy and macro-F1 of 0.654, outperforming all classical methods and approaching (within 4 points) the BERT-base primary result , while training only 4 million of 8 billion parameters. This demonstrates the remarkable parameter efficiency of LoRA adapters: by injecting low-rank matrices into the query and value projections of the pre-trained attention layers, the model can specialize to the classification task without catastrophic forgetting of pre-training knowledge.

E. Overall Comparison

Figure 12 presents a unified comparison across all experimental conditions, ranging from classical methods to large language models. The results confirm a clear performance hierarchy: (1) classical feedforward networks with static embeddings (55–64%); (2) BERT-based fine-tuning (69–73%); (3) parameter-efficient LLM fine-tuning (QLoRA, 66%). The few-shot LLM regime sits at the level of classical methods (48–60%), underscoring that in-context learning cannot substitute for gradient-based fine-tuning on a 20-class problem with an 8B-parameter model.

FIG. 12

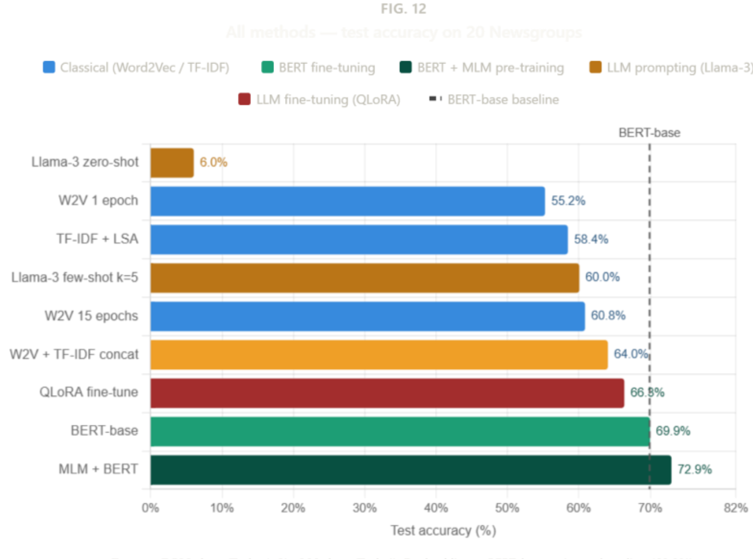

Fig. 6. Comparative bar chart of test accuracy across all experimental configurations from all four tasks. Methods are color-coded by paradigm: classical (blue), BERT-based (green), LLM prompting (orange), LLM fine-tuning (red). The dashed horizontal line marks the BERT-base primary baseline.

The performance gap between MLM+BERT (72.93%) and QLoRA fine-tuning (66.33%) is noteworthy: despite Llama-3 having 70× more parameters than BERT-base, supervised fine-tuning of BERT achieves higher accuracy on this task. This reflects both the smaller fine-tuning data size available to QLoRA (900 documents vs. 11,314 for BERT) and the task-specific architecture advantage of BERT's classification head trained end-to-end.

VI. CONCLUSION

This paper has presented a systematic comparison of five NLP paradigms on the 20 Newsgroups text classification benchmark. Our key findings are as follows.

First, Word2Vec embedding quality, as controlled by training epochs, has a direct and significant effect on downstream classification accuracy: 15 epochs yield a 10% relative improvement over 1 epoch. The t-SNE visualizations provide intuitive geometric evidence for this effect.

Second, our custom experiment (Task 1d) demonstrates that concatenating Word2Vec and TF-IDF representations into a 200-dimensional feature space yields the best classical performance (64%), confirming that semantic and discriminative signals are complementary and jointly informative.

Third, BERT fine-tuning achieves a 9+ percentage point improvement over the best classical method, with domain-adaptive MLM pre-training providing an additional 3-point gain at the cost of approximately doubling total training time. This DAPT effect is most pronounced for categories with specialized USENET-era vocabulary.

Fourth, Llama-3-8B zero-shot performance is surprisingly low (6%), primarily due to output format non-compliance. Few-shot demonstrations substantially mitigate this by providing format guidance, with k=1 alone lifting accuracy to 48%. QLoRA fine-tuning achieves 66%, bridging most of the gap to BERT-base while training fewer than 0.05% of Llama-3's parameters.

Future work should explore: (1) retrieval-augmented few-shot demonstration selection for Llama-3; (2) whole-word masking in the MLM phase; (3) multi-task learning combining MLM and classification objectives; and (4) systematic calibration of Llama-3 prompts to reduce unparseable outputs.

ACKNOWLEDGMENT

All experiments were conducted in Google Colab (T4 GPU) and tracked via Weights & Biases. AI assistance (Anthropic Claude) was used for code scaffolding; all design decisions, hyperparameter choices, and analysis are the author's own.

https://github.com/Likhita1107/NALAPRO_20Newsgroups is the link for github.

REFERENCES

[1] K. Lang, "Newsweeder: Learning to filter netnews," in Proc. 12th Int. Conf. Machine Learning (ICML), 1995, pp. 331–339.

[2] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in Advances in Neural Information Processing Systems 26 (NeurIPS), 2013, pp. 3111–3119.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems 30 (NeurIPS), 2017.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. 2019 Conf. North American Chapter of the ACL (NAACL-HLT), 2019, pp. 4171–4186.

[5] Meta AI, "Llama 3: Open foundation and fine-tuned chat models," Meta AI Research Technical Report, 2024.

[6] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Information Processing & Management, vol. 24, no. 5, pp. 513–523, 1988.

[7] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," Journal of the American Society for Information Science, vol. 41, no. 6, pp. 391–407, 1990.

[8] Q. V. Le and T. Mikolov, "Distributed representations of sentences and documents," in Proc. 31st Int. Conf. Machine Learning (ICML), 2014, pp. 1188–1196.

[9] S. Gururangan, A. Marasovic, S. Swayamditta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith, "Don't stop pretraining: Adapt language models to domains and tasks," in Proc. 58th Annual Meeting of the ACL, 2020, pp. 8342–8360.

[10] T. B. Brown et al., "Language models are few-shot learners," in Advances in Neural Information Processing Systems 33 (NeurIPS), 2020.

[11] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in Advances in Neural Information Processing Systems 36 (NeurIPS), 2023.

[12] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in Proc. 10th Int. Conf. Learning Representations (ICLR), 2022.

APPENDIX , WEIGHTS & BIASES EXPERIMENT DASHBOARDS

All experimental runs are logged to Weights & Biases under the project nalapro-20newsgroups. The following links provide direct access to training curves, metric tables, and artifact logs for each task:

Task 1 (FFNN + W2V/TF-IDF/Concat): https://wandb.ai/likhita-kolli-hochschule-luzern/nalapro-task1

Task 2 (BERT fine-tuning ablations): https://wandb.ai/likhita-kolli-hochschule-luzern/nalapro-task2-bert

Task 3 (MLM + Classification): https://wandb.ai/likhita-kolli-hochschule-luzern/nalapro-task3

Task 4 + QLoRA Bonus: https://wandb.ai/likhita-kolli-hochschule-luzern/nalapro-task4-llama3